

Memo: PPE compliance detection with RT-DETR

Repo: <https://github.com/Chan-dev12/rtdestr-vision-api> · **Weights:** release `v1.0`, `best.pt` (sha256 52019c226834)

1. Domain and data

Problem. Answer the question a site supervisor actually asks of a photo: *is anyone missing a helmet, gloves or a vest?* Helmet, safety vest, gloves, head and hands are not COCO classes.

Dataset. SH17 v1 (Kaggle, Ahmad & Rahimi 2024, CC BY-NC-SA 4.0): 8,093 Pexels photos of industrial and construction work with 17 classes. I kept the 6 needed for compliance and dropped the other 11. Compliance needs each PPE item *and* the body part it protects, so a bare head is visible as a head without a matching helmet. Tiny items such as ear-muffs and glasses would add classes the questions don't use.

Labelling. No self-labelling. The official `sh17.yaml` class order differs from the SH17 README, so I mapped ids from the yaml and checked 12 drawn previews (`reports/label_preview/`). Cleaning clipped 1,836 boxes that ran past the image edge. There were no exact duplicates.

2. Split strategy

The **test split is SH17's official `val_files.txt`** (1,620 images), unchanged, so results are comparable with the dataset authors'. The rest is split 85/15 into train (5,502) and val (971) **by near-duplicate group** (perceptual hash within 6 of 64 bits), stratified on each group's rarest class, seed 42. Grouping matters because a near-copy in train and test measures memory, not generalisation. 6 groups spanned the official split, so I dropped those 6 training images instead of moving test images. Class balance is very skewed: helmet is 2.1% of training boxes and safety vest 1.0%.

What the test set is **not**: every split comes from the same Pexels photographers and style. It is stock photography, not fixed site-camera footage.

3. Evaluation

Test split (1,620 img)	mAP50	mAP50-95	P @0.65	R @0.65
all classes	0.781	0.560	0.922	0.832
helmet / safety-vest / gloves	0.761 / 0.529 / 0.667	0.555 / 0.323 / 0.400	0.78 / 0.75 / 0.84	0.68 / 0.37 / 0.51

- **Per class.** Person, head and hands have recall of 0.82 to 0.89. The PPE classes that matter most are the weakest, in line with their share of training data.
- **Threshold 0.65** is the F1-optimal point of the sweep. Moving from 0.5 to 0.65 traded recall (0.865 to 0.832) for precision (0.872 to 0.922).
- **By size.** Recall is 0.94 for large boxes, 0.80 for medium, 0.29 for small. Small boxes cause 694 of the 1,541 misses.
- **Confusion.** Almost all errors are against background: 1,345 missed boxes and 366 false positives. Of 40 true class swaps, 24 are gloves and hands, and 8 are helmet and head.

What these numbers don't tell you. (1) The threshold was picked on the test split, so P and R at 0.65 are slightly optimistic; validation should have been used. (2) Test shares source and style with train, so this is an in-distribution upper bound. I did not build an out-of-source set. (3) The micro-averaged P and R are dominated by person, head and hands (91% of boxes), which hides the 0.37 vest recall. (4) mAP says nothing about whether a *compliance answer* is right; one missed helmet flips an image-level answer. (5) mAP50-95 penalises loose boxes that don't affect counting.

4. Five failure cases

Images are in `reports/test/failures/`: ground truth green, predictions red, misses yellow. The counts come from `errors.csv`.

#	Image	What went wrong	Root cause and evidence	Fix
1	<code>01_... efc3b6dea9f6</code>	54 boxes missed on a truck full of workers	Crowding + top-down viewpoint. Drone view, bodies overlap. 9 misses typed <code>missed_crowded</code> ; 29 missed hands and 10 missed heads are small.	Aerial/overhead images; tiled inference
2	<code>03_... 7fb5a31071bf</code>	All 34 people and heads on a ferry deck missed, zero predictions	Small objects. All 34 boxes are under 32x32 px at 1280 px, then shrunk again to 640 for the model. Small-person recall is 0.14 overall.	Train/infer at 960 to 1280 px, or tiling
3	<code>15_... 9bbbb41ddd08</code>	A glove predicted as hands (0.78)	Class confusion, gloves vs hands. The glove box is small and low-contrast. This is the largest swap: 16 gloves labelled hands, 8 the reverse. Gloves recall 0.51.	More close-up glove data; a crop classifier on hand boxes

#	Image	What went wrong	Root cause and evidence	Fix
4	19_... 644ccc946e33	Night street: a cyclist missed, a dark helmet boxed as head (0.72)	Low light. Brightness about 0.20, the darkest of the 30 worst images. The dark helmet has little contrast with the scene. For /ask this produces a <i>false</i> "no helmet". The R3 dark flag (below 0.18) does not fire.	Brightness augmentation; raise the dark threshold to about 0.25
5	27_... 6e7d81ee5a3f	4 head false positives (0.76 to 0.87) on women winnowing rice	Label noise. The heads are clearly visible, covered by scarves, but ground truth has no head box for them. The model is right. Reported head precision is therefore pessimistic.	Relabel; audit false positives above 0.75

Pattern. Missing objects dominates, and size is the strongest single cause. With two more days I would train at 960 px and build a small out-of-source set of site-camera frames to measure the real gap.

5. Part B: when the detector is called

The router maps each question to one intent. `meta` (what can you detect), `general` (unrelated) and `unsupported` (colours, identities, text, unknown objects) are answered **without** RT-DETR. Everything else runs the detector. Python then turns boxes into counts, 3x3 regions, image-quality flags, and one-to-one body-part to PPE matches, so the LLM never counts. Seven deterministic rules (`policy.py`) decide whether those facts support an answer. The key one, R3, says the absence of a detection is evidence only if the image is usable and the class's measured test recall is at least 0.60. An LLM rewording that introduces a number not in the facts is rejected.

"Insufficient information" example (real output, `docs/samples/ask_hard_image.json`):

Q: "Are there any safety vests?" on a darkened copy of `docs/samples/bus.jpg`

Facts: 4 people and 0 vests detected; image flagged "very dark"; vest test recall 0.37.

A: "I didn't detect any safety vests, but I can't be confident there are none: the image looks very dark, which is a known cause of missed detections; the detector only finds 37% of safety vests on its test set." (status: `insufficient_information`, rule R3)

Answering "no" would be a guess. The detector misses 63% of vests even on clean test images.